

- (1) (25 pts.) Consider a mixture of gasses consisting of three species, namely, A, B, and C. It is known that the mixture is made up of 40 mol% A, 18.75 wt % of B, 20 mol% of C. The molecular weight of A is 40 kg/kmole and of C is 50 kg/kmole.

(a) Calculate the molecular weight of B in kg/kmole. (15 pts.)

Basis: 100 kmol total

Component	mol%	$M (\frac{\text{kg}}{\text{kmole}})$	$m (\text{kg})$
A	40	40	1600
B	40	x	$40x$
C	20	50	1000
	<u>100</u>		<u>$2600 + 40x$</u>

$$\text{wt}\%: 0.1875 = \frac{40x}{2600 + 40x}$$

$$40x = 487.5 + (0.1875)40x$$

$$40(1 - 0.1875)x = 487.5$$

$$x = \frac{487.5}{40(1 - 0.1875)} = 15 \text{ kg/kmole}$$

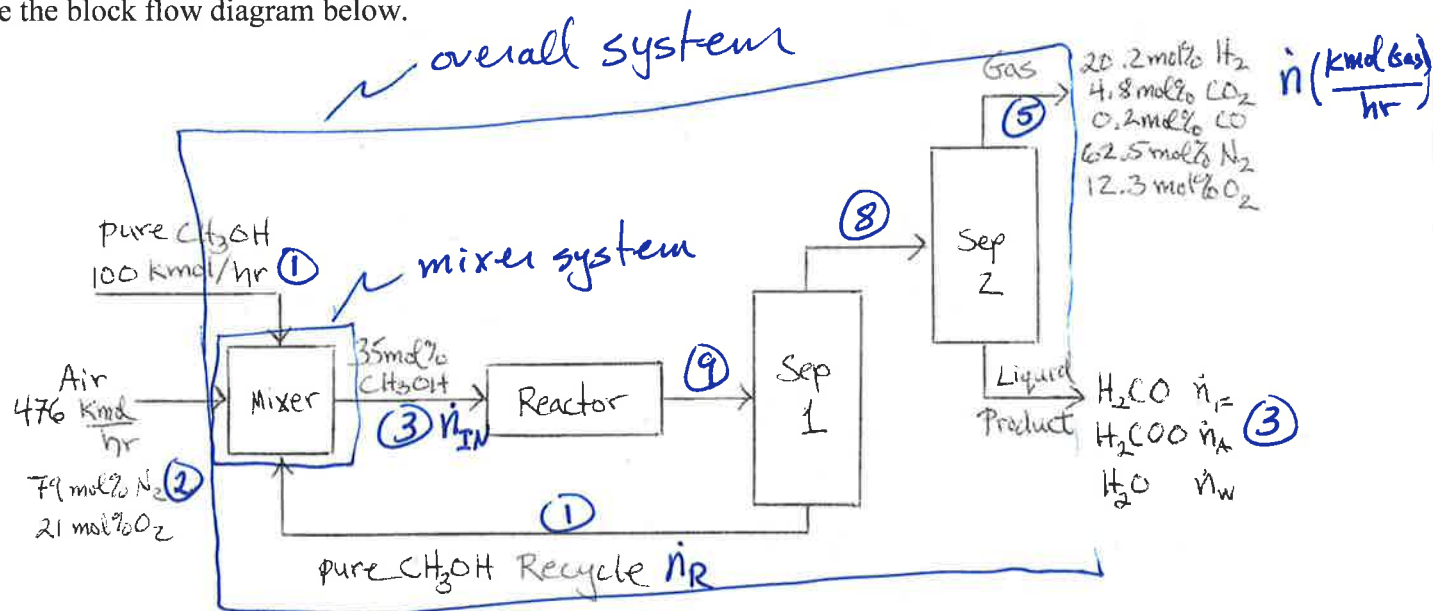
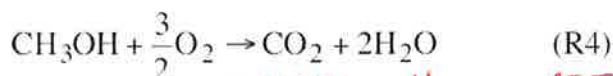
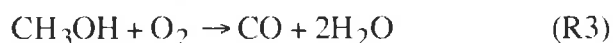
$$M_B = \underline{15} \text{ kg/kmole}$$

- (b) Calculate the mole average molecular weight of the mixture kg/kmole. (10 pts.)

$$\begin{aligned} \bar{M} &= \sum_i x_i M_i = (0.4)(40) + (0.4)15 + (0.2)50 \\ &= 32 \text{ kg/kmole} \end{aligned}$$

$$\bar{M} = \underline{32} \text{ kg/kmole}$$

(2) (50 pts.) Formaldehyde (H_2CO) is produced by oxidation of methanol (CH_3OH). As shown, four reactions take place in the reactor. Pure methanol (CH_3OH) at 100 kmol/hr along with 476 kmol/hr of air (21 mol% O_2 and 79 mol% N_2) and a recycle stream of pure methanol are mixed and then enter the reactor. The reactor inlet stream is 35 mol% CH_3OH . The reactor exit stream goes to a separator with one exit stream of pure methanol that is recycled to the mixer and another stream that goes to a second separator that has an exit stream of gas with 20.2 mol% H_2 , 4.8 mol% CO_2 , 0.2 mol% CO , 62.5 mol% N_2 , and 12.3 mol% O_2 and a liquid product stream containing formaldehyde (H_2CO), formic acid (H_2COO), and water (H_2O). See the block flow diagram below.



(a) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Fill in the table below giving a brief explanation for your entries. Compute the DOF. (10 pts.)

	Number	Explanation
4a. Stream Variables	32	$1+2+3+9+8+1+5+3$
4b. System Variable	4	four Reactions given
Total Independent Variables	36	
5a. Specified Flows	2	CH_3OH and air into mixer
5b. Specified Stream Compositions	6	① air in + ④ gas extst + ① CH_3OH into Rx
5c. Specified System Performances	0	none given
5d. Material Balance Equations	29	③ mixer + ⑨ reactor + ⑨ Sep 1 + ⑧ Sep 2
Total Independent Equations	37	
DOF	-1	overspecified*

* Missing one chemical reaction equation for production of H_2 !

PROBLEM 2 CONTINUED

- (b) Calculate the product stream molar flow rates in kmol/hr for the formaldehyde (H_2CO) $\dot{n}_F$, formic acid (H_2COO) $\dot{n}_A$, and water (H_2O) $\dot{n}_W$ in kmol/hr. Hint: Consider an atomic balance. (30 pts.)

Atomic Balance on overall system

$$\text{C: } 100 \frac{\text{kmol}}{\text{hr}} = [(0.048) + (0.002)] \dot{n} + \dot{n}_F + \dot{n}_A$$

$$\text{H: } 400 \frac{\text{kmol}}{\text{hr}} = [2(0.202)] \dot{n} + 2\dot{n}_F + 2\dot{n}_A + 2\dot{n}_W$$

$$\text{O: } [100 + 2(0.21)476] \frac{\text{kmol}}{\text{hr}} = [2(0.048) + (0.002) + 2(0.123)] \dot{n} + \dot{n}_F + 2\dot{n}_A + \dot{n}_W$$

$$\text{N: } 2(0.79)476 \frac{\text{kmol}}{\text{hr}} = 2(0.625) \dot{n} \Rightarrow \dot{n} = \frac{(0.79)(476)}{(0.625)} = 601.66 \frac{\text{kmol}}{\text{hr}}$$

$$\text{C: } 69.917 = \dot{n}_F + \dot{n}_A$$

$$\text{H: } 156.93 = 2\dot{n}_F + 2\dot{n}_A + 2\dot{n}_W \Rightarrow 156.93 = 2\dot{n}_F + 2(69.917 - \dot{n}_F) + 2\dot{n}_W$$

$$\text{O: } 92.948 = \dot{n}_F + 2\dot{n}_A + \dot{n}_W \quad \dot{n}_W = \frac{156.93 - 2(69.917)}{2}$$

$$92.948 = \dot{n}_F + 2(69.917 - \dot{n}_F) + 8.5471 = 8.5471$$

$$\dot{n}_F = 55.4331$$

$$\dot{n}_A = 69.917 - 55.4331 = 14.484$$

$$\dot{n}_F = \underline{55.4331} \text{ kmol H}_2\text{CO /min}, \quad \dot{n}_A = \underline{14.484} \text{ kmol H}_2\text{COO /min}, \quad \dot{n}_W = \underline{8.5471} \text{ kmol H}_2\text{O/min}$$

- (c) Calculate the fractional conversion of methanol for the reactor (the single-pass conversion). (10 pts.)

mixer balance

$$\text{Total moles: } 100 \frac{\text{kmol}}{\text{hr}} + 476 \frac{\text{kmol}}{\text{hr}} + \dot{n}_R = \dot{n}_I$$

$$\text{CH}_3\text{OH moles: } 100 \frac{\text{kmol}}{\text{hr}} + \dot{n}_D = 0.35 \dot{n}_I$$

$$476 = 0.65 \dot{n}_I \Rightarrow \dot{n}_I = 732.3$$

$$f_{\text{CH}_3\text{OH}} = \frac{(0.35)732.3 - 100}{(0.35)(732.3)} = \frac{100}{256.3} = 0.39$$

$$f_{\text{C, methanol}} = \underline{0.39}$$

Selectivity =

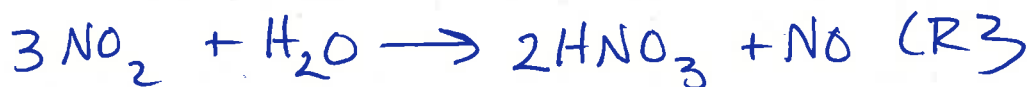
(25 pts.) The Ostwald Process produces nitric acid (HNO_3) by the oxidation of ammonia (NH_3). The process consists of three reactions:

R1: Ammonia reacts with oxygen (O_2) to form nitric oxide (NO) and water (H_2O).

R2: NO reacts with O_2 to form nitrogen dioxide (NO_2).

R3: NO_2 is bubbled through water to produce HNO_3 and NO .

- (a) Write a balanced chemical reaction equation for each of these three reactions. Make all of the stoichiometric coefficients the smallest whole numbers possible. (9 pts.)



- (b) Use a generation-consumption analysis to synthesize a reaction pathway to make nitric acid from ammonia and oxygen based on the three reactions in part (a) with no net generation or consumption of NO or NO_2 . Water is allowed as a byproduct. (9 pts.)

	R1 $x_1=1$	R2 $x_2=3$	R3 $x_3=2$	Overall Rxn
NH_3	-4	0	0	-4
O_2	-5	-1	0	-8
NO	+4	-2	+1	0
NO_2	0	+2	-3	0
H_2O	+6	0	-1	+4
HNO_3	0	0	+2	+4

Write the net chemical equation for the process.

$$\begin{aligned} \text{NO: } 4x_1 - 2x_2 + x_3 &= 0 & \text{Try } x_1=1 & \quad -2x_2 + x_3 = -4 \\ \text{NO}_2: 2x_2 - 3x_3 &= 0 & & \quad 2x_2 - 3x_3 = 0 \end{aligned}$$

$$\begin{aligned} -2x_3 &= -4 \\ x_3 &= 2 \Rightarrow x_2=3 \end{aligned}$$



- (c) Calculate the grams of NH_3 and grams of O_2 required to make 100 grams of nitric acid. Atomic Weights: $\text{H} = 1 \text{ g/mol}$, $\text{N} = 14 \text{ g/mol}$, and $\text{O} = 16 \text{ g/mol}$. (7 pts.)



$$M_{\text{HNO}_3} = 1 + 14 + 3(16) = 63 \text{ g/mol}$$

$$M_{\text{NH}_3} = 14 + 3 = 17 \text{ g/mol}$$

$$\begin{aligned} \frac{100 \text{ g HNO}_3}{63 \text{ g/mol}} &= 1.5873 \text{ mol} \\ M_{\text{NH}_3} &= \left(\frac{1 \text{ mol NH}_3}{1 \text{ mol HNO}_3} \right) (1.5873 \text{ mol}) (17 \text{ g/mol}) \\ &= 26.984 \end{aligned}$$

$$\text{Mass of O}_2 = 101.59 \text{ g O}_2$$

$$\text{Mass of NH}_3 = 26.984 \text{ g NH}_3$$

$$m_{\text{O}_2} = \left(\frac{2 \text{ mol O}_2}{1 \text{ mol HNO}_3} \right) (1.5873 \text{ mol HNO}_3) \left(32 \frac{\text{g O}_2}{\text{mol O}_2} \right) = 101.59 \text{ g O}_2$$

2b overall balance

$$\dot{n}_{\text{CH}_3\text{OH}, \text{out}} = \dot{n}_{\text{CH}_3\text{OH}, \text{in}} - \dot{\xi}_1 - \dot{\xi}_2 - \dot{\xi}_3 - \dot{\xi}_4 - \dot{\xi}_5$$

$$\dot{n}_{\text{O}_2, \text{out}} = \dot{n}_{\text{O}_2, \text{in}} - \frac{1}{2}\dot{\xi}_1 - \dot{\xi}_2 - \dot{\xi}_3 - \frac{3}{2}\dot{\xi}_4$$

$$\dot{n}_{\text{N}_2, \text{out}} = \dot{n}_{\text{N}_2, \text{in}} \Rightarrow (0.79)476 \frac{\text{kmol}}{\text{hr}} = (0.625)\dot{n} \Rightarrow \dot{n} = 601.66 \frac{\text{kmol}}{\text{hr}}$$

$$\dot{n}_{\text{H}_2, \text{out}} = \dot{n}_{\text{H}_2, \text{in}} + \dot{\xi}_5 \Rightarrow \dot{\xi}_5 = (0.202)\dot{n} = 121.535$$

$$\dot{n}_{\text{CO}_2, \text{out}} = \dot{n}_{\text{CO}_2, \text{in}} + \dot{\xi}_4 \Rightarrow \dot{\xi}_4 = (0.048)\dot{n} = 28.8797$$

$$\dot{n}_{\text{CO}, \text{out}} = \dot{n}_{\text{CO}, \text{in}} + \dot{\xi}_3 \Rightarrow \dot{\xi}_3 = (0.002)\dot{n} = 1.20332$$

$$\dot{n}_F = \dot{n}_{\text{H}_2\text{CO}, \text{out}} = \dot{n}_{\text{H}_2\text{CO}, \text{in}} + \dot{\xi}_1 + \dot{\xi}_5 = 55.435$$

$$\dot{n}_A = \dot{n}_{\text{H}_2\text{CO}_2, \text{out}} = \dot{n}_{\text{H}_2\text{CO}_2, \text{in}} + \dot{\xi}_2 \Rightarrow \dot{\xi}_2 = 14.484$$

$$\dot{n}_W = \dot{n}_{\text{H}_2\text{O}, \text{out}} = \dot{n}_{\text{H}_2\text{O}, \text{in}} + \dot{\xi}_1 + \dot{\xi}_2 + 2\dot{\xi}_3 + 2\dot{\xi}_4 = 8.55$$

$$\text{total } \dot{n}_F + \dot{n}_A + \dot{n}_W + \dot{n} = 576 \frac{\text{mol}}{\text{hr}} + \frac{1}{2}\dot{\xi}_1 + \dot{\xi}_3 + \frac{1}{2}\dot{\xi}_4 + \dot{\xi}_5$$

$$\begin{aligned} 100 \frac{\text{kmol}}{\text{hr}} &= \dot{\xi}_1 + \dot{\xi}_2 + 1.20332 + 28.8797 + 121.535 = \\ &= \dot{\xi}_1 + \dot{\xi}_2 + ((0.202) + (0.048) + (0.002))(601.66) \end{aligned}$$

$$100 = \dot{\xi}_1 + \dot{\xi}_2 + 51.61832$$

$$(0.21)476 = (0.123)601.66 + \frac{1}{2}\dot{\xi}_1 + \dot{\xi}_2 + 1.20332 + \frac{3}{2}(28.8797)$$

$$-18.567 = \frac{1}{2}\dot{\xi}_1 + \dot{\xi}_2$$

$$-51.61832 = \dot{\xi}_1 + \dot{\xi}_2$$

$$\boxed{\dot{\xi}_1 = -66.10} \quad \dot{\xi}_2 = +14.484$$

1st reaction runs backwards!

